Questions

1. A particular system is controlled by an operator through commands entered from a keyboard. The average number of commands entered in an 8-hour interval is 60. Suppose the processor scans the keyboard every 100 ms. How many times will the keyboard be checked in an 8-hour period? By what fraction would the number of processor visit to the keyboard be reduced if interrupt-driven I/O were used?

2. A microprocessor scans the status of an output I/O device every 20 ms. This is accomplished by means of a timer alerting the processor every 20 ms. The interface of the device includes two ports: one for status and one for data output. How long does it take to scan and service the device given a clocking rate of 8 MHz? Assume for simplicity that all instruction cycles take 12 clock cycles.

3. Consider a microprocessor having a block I/O transfer instruction. Following its first execution, such an instruction takes five clock cycles to re-execute and 20 clock cycles of initial fetching. However, if we employ a non-blocking I/O instruction, it takes a total of 20 clock cycles for fetching and execution. Calculate the increase in speed with the block I/O instruction when transferring blocks of 128 bytes.

4. For programmed I/O, the processor is stuck in a wait loop doing status checking of an I/O device. To increase efficiency, the I/O software could be written so that the processor periodically checks the status of the device. If the device is not ready, the processor can jump to other tasks. After some timed interval, the processor comes back to check status again. Consider the above scheme for outputting data one character at a time to a printer that operates at 10 characters per second (cps). What will happen if its status is scanned every 200 ms? Next consider a keyboard with a single character buffer. On average, characters are entered at a rate of 10 cps. However, the time interval between two consecutive key depressions can be as short as 60 ms. At what frequency should the keyboard be scanned by the I/O program?

5. A computer consists of a processor and an I/O device D connected to main memory M via a shared bus with a data bus width of one word. The processor can execute a maximum of 10^6 instructions per second. An average instruction requires five machine cycles, three of which use the memory bus. A memory read or write operation uses one machine cycle. Suppose that the processor is continuously executing "background" programs that require 95% of its instruction execution rate but not any I/O instructions. Assume that one processor cycle equals one bus cycle. Now suppose the I/O device is to be used to transfer very large blocks of data between M and D.
   a) If programmed I/O is used and each one-word I/O transfer requires the processor to execute two instructions, estimate the maximum I/O data-transfer rate, in words per second, possible through D.
   b) Estimate the same rate if DMA is used.

6. A 32-bit computer has two selector channels and one multiplexor channel. Each selector channel supports two magnetic disk and two magnetic tape units. The multiplexor channel has two line printers, two card readers, and 10 VDT terminals connected to it.
   Assume the following transfer rates:
   - Disk drive 800 Kbytes/s
   - Magnetic tape drive 200 Kbytes/s
   - Line printer 6.6 Kbytes/s
   - Card reader 1.2 Kbytes/s
   - VDT 1 Kbyte/s

   Estimate the maximum aggregate I/O transfer rate in this system.